The Impossibility of Faithful Explanation Under Underdetermination

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Abstract

When a system is underspecified—when multiple configurations produce the same observable output—any explanation faces an inherent tradeoff. We prove that no explanation can simultaneously be faithful (consistent with the system's native explanation), stable (invariant across observationally equivalent configurations), and decisive (committing to every distinction the native explanation makes). This impossibility holds whenever the Rashomon property obtains: equivalent configurations with incompatible explanations. The Rashomon property is sufficient for impossibility (Theorem 1) and necessary for fully specified systems (Proposition 3). For systems with non-injective observations and compatible-but-distinct explanations, the relationship is more nuanced. The theorem is instantiated across nine domains: feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery, model selection, saliency maps, token citation explanations, and mechanistic interpretability. For causal discovery, the Rashomon property is derived from first principles (Markov equivalence). The resolution—G-invariant aggregation—is provably stable among explanation methods. The result is mechanically verified in the Lean 4 proof assistant (349 theorems, 0 unproved goals).

Significance. Explainability is central to trustworthy AI, scientific inference, and regulatory compliance. We prove a universal impossibility: no method of explaining an underspecified system can simultaneously be faithful, stable across equivalent configurations, and decisive. This formalizes the explanatory consequences of contrastive underdetermination [1]. Prior formal analyses of underdetermination in Bayesian frameworks [2, 3] did not yield impossibility results for explanation methods. The impossibility applies to nine explanation paradigms in machine learning—from SHAP values to circuit analysis—and extends to scientific explanation and medical diagnosis. The constructive resolution (aggregate over equivalent configurations) provides a provably stable tradeoff. The entire proof is mechanically verified.

Keywords: impossibility theorem, explainability, underdetermination, Rashomon effect, formal verification

1 Introduction

When multiple configurations of a system produce identical observable output, any attempt to explain the system's behavior faces a fundamental constraint. This paper proves that constraint is an impossibility: no explanation can simultaneously be faithful, stable, and decisive.

This structure arises across science: Quine–Duhem underdetermination [4, 5], neural degeneracy [6, 7], the Rashomon effect [8, 9], partial identification [10], and the view update problem [11].

We prove the impossibility, characterize the Rashomon property as sufficient for impossibility and necessary for fully specified systems, demonstrate it across nine domains with the most detailed treatment in machine learning, and provide a constructive resolution. Informally:

> *If two configurations of a system produce the same observable output but honestly report incompatible explanations, then no explanation method can simultaneously be faithful to the system, stable across equivalent configurations, and decisive in its commitments.*

The proof is four steps—decisiveness, stability, faithfulness, contradiction—and is mechanically verified in the Lean 4 proof assistant [12] (74 files, 349 theorems, 72 axioms, 0 `sorry`). The core impossibility requires zero axioms beyond the Rashomon property itself.

Our contributions are: (1) To our knowledge, this is a universal impossibility theorem unifying nine domain-specific results under a single proof; (2) a proof that the Rashomon property is sufficient for impossibility and necessary for fully specified systems; (3) nine instances spanning feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery (with Rashomon *derived* from Markov equivalence), model selection, saliency maps, token citation explanations, and mechanistic interpretability; (4) a constructive resolution via G-invariant aggregation proved uniquely optimal; (5) em-

pirical validation across seven instances; and (6) complete mechanical verification. The resolution strategy is structurally analogous to invariant decision procedures in the Hunt–Stein framework [13].

2 Framework

2.1 Explanation Systems

We introduce a minimal abstract framework in which the impossibility can be stated and proved. All nine instances in Section 3 are special cases.

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{observe}, \mathsf{explain}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* configuration space*—the set of possible configurations of the system (e.g., model parameters, experimental setups, circuit wirings);*
- $\mathcal{H}$ *is an* explanation space*—the set of possible explanations;*
- Y *is an* observable space*—the set of measurable outputs;*
- $\mathsf{observe} : \mathbf{\Theta} \to Y$ *assigns to each configuration its observable behavior;*
- $\mathsf{explain} : \mathbf{\Theta} \to \mathcal{H}$ *assigns to each configuration the explanation it "honestly reports"; and*
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ *is an* incompatibility relation *satisfying* irreflexivity*:* $\neg \perp (h, h)$ *for every* $h \in \mathcal{H}$*.*

An *explanation map* $E : \mathbf{\Theta} \to \mathcal{H}$ is a procedure that a practitioner *chooses*. The native explanation $\mathsf{explain}(\theta)$ records what the configuration actually implies; $E(\theta)$ is what the chosen method reports.

2.2 Three Desiderata

Definition 2 (Faithful, Stable, Decisive)**.** *An explanation map* E *is:*

- ***Faithful*** *if* $E(\theta)$ *never contradicts the native explanation:* $\forall\, \theta,\ \neg \perp (E(\theta), \mathsf{explain}(\theta))$*.*
- ***Stable*** *if equivalent configurations receive the same explanation:* $\forall\, \theta_1, \theta_2,\ \mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \Rightarrow E(\theta_1) = E(\theta_2)$*.*
- ***Decisive*** *if* $E(\theta)$ *inherits every incompatibility of the native explanation:* $\forall\, \theta, h,\ \perp (\mathsf{explain}(\theta), h) \Rightarrow \perp (E(\theta), h)$*.*

Faithfulness says the method does not contradict the system. Stability says the method is reproducible—it does not depend on which equivalent configuration happens to be examined. Decisiveness says the method is at least as committal as the system itself: whatever the native explanation rules out, the chosen explanation also rules out.

2.3 The Rashomon Property

Definition 3 (Rashomon Property)**.** *An explanation system has the* Rashomon property *if there exist* $\theta_1, \theta_2 \in \mathbf{\Theta}$ *with*

$$\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \quad \textit{and} \quad \perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2)).$$

Two configurations produce the same observable output yet honestly report incompatible explanations. This is the formal statement of underdetermination at the explanation level.

2.4 Main Theorem

Theorem 1 (Universal Explanation Impossibility)**.** *If an explanation system has the Rashomon property, then no explanation map* E *can simultaneously be faithful, stable, and decisive.*

Proof. Let θ_1, θ_2 witness the Rashomon property. Suppose E is faithful, stable, and decisive.

1. **Decisiveness** at θ_1: $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ implies $\perp (E(\theta_1), \mathsf{explain}(\theta_2))$.
2. **Stability**: $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$, so $\perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
3. **Faithfulness** at θ_2: $\neg \perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
4. Steps 2 and 3 contradict. □

Proposition 1 (Tightness)**.** *Each pair of properties is simultaneously achievable: (i) faithful + decisive (set* $E = \mathsf{explain}$*; stability fails); (ii) faithful + stable (set* E *to a neutral constant; decisiveness fails); (iii) stable + decisive (set* E *to a maximally committal constant; faithfulness fails).*

Proposition 2 (Necessity)**.** *An explanation system admits a faithful, stable, and decisive explanation map if and only if it does* not *have the Rashomon property.*

The Rashomon property is sufficient for impossibility (Theorem 1) and necessary for fully specified systems: when each observation uniquely determines its configuration, $E = \mathsf{explain}$ satisfies all three properties (Proposition 2). For systems with non-injective observations and compatible-but-distinct explanations on the same fiber, the relationship between Rashomon and impossibility is more nuanced (see *SI Appendix*). The proofs of Propositions 1–2 are in *SI Appendix*.

The simplicity of the proof—four steps from the Rashomon property to contradiction—is a feature of the framework, not a limitation of the result. The intellectual content lies in identifying the right definitions: faithful as non-contradiction, stable as factoring through observations, decisive as inheriting incompatibilities. The axiom substitution analysis (*SI Appendix*, Section S4) shows these definitions are uniquely calibrated: strengthening decisive collapses the trilemma to a dilemma; weakening it loses the impossibility entirely. The contribution is the axiomatic framework and its unification of nine domains, not the proof technique.

3 Instances

We instantiate Theorem 1 in four domains chosen for cross-disciplinary breadth. Five additional instances (attention maps, counterfactual explanations, concept probes, model selection, language model explanations) are detailed in *SI Appendix*.

3.1 Feature Attribution (Machine Learning)

The 6-tuple: $\mathbf{\Theta}$ = trained models (GBDT, Lasso, neural networks); Y = input–output function; $\mathcal{H}$ = attribution vectors $(\varphi_1, \ldots, \varphi_P)$; $\perp$ = opposite ranking of a collinear pair. Collinearity induces a Rashomon set of near-optimal models differing only in which correlated feature acts as primary predictor [14, 15]. The impossibility follows from Theorem 1, subsuming [16] as a special case.

Quantitatively: for GBDTs the attribution ratio diverges as $1/(1-\rho^2)$; for Lasso one feature is zeroed entirely (infinite ratio). Empirically, SHAP rankings flip at a 33% rate across equivalent models.

3.2 Causal Discovery

The 6-tuple: $\mathbf{\Theta}$ = DAGs over vertex set V; Y = conditional independence (CI) structures; $\mathsf{observe}$ = d-separation Markov condition; $\mathcal{H}$ = fully oriented DAGs; $\perp$ = opposite orientation of some edge. Two DAGs are *Markov equivalent* iff they encode the same CI structure [17].

Proposition 3 (Causal Rashomon Property—Derived)**.** *Any CPDAG with an undirected edge $\{X, Y\}$ witnesses the Rashomon property: orient $X \to Y$ for h and $Y \to X$ for h'; both are Markov-consistent yet $\perp(h, h')$.*

By Theorem 1, no orientation rule can simultaneously be faithful, stable, and decisive. The resolution is to report CPDAGs [18]—trading decisiveness for faithfulness and stability.

3.3 Mechanistic Interpretability

The 6-tuple: $\mathbf{\Theta}$ = weight configurations; Y = input–output function; $\mathcal{H}$ = circuit decompositions; $\perp$ = contradictory attribution of the same computation to different network components. [19] found **85 distinct valid circuits** for a single XOR task; [20] show sparse autoencoders learn different features per seed.

By Theorem 1, no circuit explanation can simultaneously be faithful, stable, and decisive. The resolution is to report the circuit equivalence class—features invariant across all valid decompositions form the "skeleton"; decomposition-dependent features are the "reversible edges." For AI safety: arguments that a specific circuit is THE mechanism of a capability must account for alternative circuits with potentially different safety properties.

Table 1: The three escape routes from the explanation trilemma.

Drop	Lose	Gain	Tools
Faithful	May contradict	Stable + decisive	Fixed rules
Stable	Changes on retrain	Faithful + decisive	SHAP, single circui
Decisive	Ties, uncertainty	Faithful + stable	Dash, CPDAG

3.4 Neural Degeneracy (Neuroscience)

The 6-tuple: $\mathbf{\Theta}$ = neural circuit parameterizations (ionic conductances, synaptic weights); Y = motor pattern output; $\mathcal{H}$ = mechanistic accounts of which ion channels drive the rhythm; $\perp$ = contradictory primary-channel attributions. Marder and Goaillard [6] documented that crustacean stomatogastric ganglia with identical motor patterns arise from qualitatively different conductance profiles—*degeneracy* [7]—which is precisely the Rashomon property.

By Theorem 1, no mechanistic account can simultaneously be faithful to the specific circuit, stable across degenerate circuits, and decisive about the primary ionic mechanism. Biological and artificial neural circuits face the same structural impossibility.

The remaining five instances—attention maps (19.9% argmax flip rate across retrained DistilBERT models[1]), counterfactual explanations (23.5% direction flip rate), concept probes (0.90 direction instability), model selection (80% best-model flip rate), and token citation explanations (34.5% citation flip rate)—are detailed with full experimental methodology in *SI Appendix*.

Of the nine instances, the causal discovery instance has a fully derived Rashomon property: Markov equivalence is proved constructively from a 3-node example in Lean (*SI Appendix*, Section S2.5). The remaining eight instances axiomatize the Rashomon property as formalized domain knowledge—empirically justified by cited literature for each domain but not derived from primitive assumptions within the proof assistant. The core impossibility theorem itself requires zero axioms; the instance-level axioms encode the domain-specific claim that observational equivalence with incompatible explanations occurs.

4 Resolution

4.1 Escape Routes

The impossibility is tight: one desideratum must be sacrificed. Table 1 summarizes the three escape routes.

[1]Weight perturbation yields 60.0%; we report the more conservative full-retraining figure (20 models, last 2 transformer layers retrained from random initialization).

4.2 G-Invariant Aggregation

Let G be a group acting on the configuration space such that $g \cdot \theta$ preserves observables: $\mathsf{observe}(g \cdot \theta) = \mathsf{observe}(\theta)$. An explanation map is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g, θ.

Proposition 4 (G-Invariance Implies Stability)**.** *If G acts transitively on observe-fibers and E is G-invariant, then E is stable.*

When G is finite and the explanation space admits averaging, the canonical G-invariant map is the *orbit average*:

$$\bar{E}(\theta) = \frac{1}{|G|} \sum_{g \in G} \mathsf{explain}(g \cdot \theta).$$

This map is stable (by G-invariance), faithful in expectation (the expected explanation matches the expected native explanation over the orbit), and provably optimal among stable maps. It sacrifices decisiveness: where the orbit disagrees, the average produces ties.

The resolution strategy—restricting to G-invariant explanation maps—is structurally analogous to the invariant decision procedures justified by the Hunt–Stein theorem (Lehmann & Romano 2005, Ch. 6). A formal minimax optimality proof for the abstract ExplanationSystem requires additional structure (a loss function on $\mathcal{H}$, an amenable group action, and a risk functional) and is a natural direction for future work. For the attribution instance, the companion paper proves Dash achieves optimal faithfulness among stable methods.

4.3 Instance-Specific Resolutions

The abstract resolution instantiates concretely: Dash (average SHAP over retrained models, with consensus equity proved in Lean 4); CPDAGs for causal discovery [18]; circuit equivalence classes for mechanistic interpretability; the degenerate set for neural circuits; and differential diagnosis for medicine.

5 Empirical Illustration

The impossibility theorem is a mathematical proof; these experiments demonstrate that the Rashomon property produces measurable instability in practice, not that the theorem depends on experimental evidence. We illustrate the impossibility empirically across seven of the nine instances (Table 2). For each instance, we train multiple models that are functionally equivalent (matched on prediction accuracy) and measure explanation instability.

Three results deserve emphasis. First, the 19.9% attention argmax flip rate under full retraining: despite 100% prediction agreement among retrained DistilBERT models on SST-2, the most-attended token changes for a substantial fraction of sentences. Second, the 0.90 concept direction instability: concept activation vectors extracted from 15 equivalent MLP classifiers are nearly orthogonal (mean $|\cos| = 0.10$), meaning the "concept direction" is almost entirely a function of the random seed, not the function the model computes. Third, the 0% instability for negative controls (instances where the Rashomon property does not hold, e.g., unique-solution linear regression), confirming that the theorem's boundary condition is empirically sharp.

Table 2: Cross-instance empirical illustration. All instability measures have 95% bootstrap CIs excluding zero. Prediction equivalence confirms functional identity.

Instance	Metric	Value	Pred. equiv.
Attribution	SHAP flip	33%[‡]	>99%
Attention	Argmax flip	19.9%[‖]	100%
Counterfactual	Dir. flip	23.5%	ΔAUC<0.03
Concept probe	$1-\lvert\cos\rvert$	0.90	97.2%
Model selection	Best flip	80%	ΔAUC<0.02
Saliency map (GradCAM)	Peak flip	9.6%	78.8%[†]
Token citation	Citation flip	34.5%	100%[§]

[†] GradCAM prediction agreement is 78.8% ($\sigma = 0.0005$); the modest instability (9.6%) may partly reflect prediction divergence rather than pure Rashomon instability.

[‡] Four of twenty models achieved 72–78% held-out accuracy; all twenty are included in the analysis.

[§] Token citation extracted via attention rollout as a proxy.

[‖] Weight perturbation yields 60.0%; we report the conservative retraining figure.

Full experimental details—datasets, model architectures, perturbation protocols, confidence intervals, and negative controls—are in *SI Appendix*.

Information-theoretic perspective. By the data processing inequality, any stable map forfeits at least $H(\mathsf{explain}(\boldsymbol{\Theta}) \mid Y)$ bits of the native explanation. This complements Theorem 1: both identify the Rashomon property as root cause, but the information-theoretic bound quantifies minimum information loss.

6 Discussion

6.1 Implications Beyond Machine Learning

Beyond ML, the theorem applies to any domain with the ExplanationSystem structure: database view updates [11], Rashomon sets in statistics [15], neural degeneracy [6, 7], partial identification in econometrics [10], and differential diagnosis in medicine. Formalizing each as an ExplanationSystem and verifying the Rashomon property yields the impossibility as a corollary.

Tier 3: Suggestive connections (discussion only). The Quine–Duhem underdetermination thesis [4, 5] is the informal precursor to our result. Theorem 1 proves an

impossibility theorem for explanation under observational equivalence—formalizing, to our knowledge for the first time, the explanatory consequences of what philosophers of science call contrastive underdetermination [1]. Prior formal analyses of underdetermination in Bayesian frameworks [2, 3] did not yield impossibility results for explanation methods. The G-invariant resolution is structurally reminiscent of epistemic structural realism [21]: stable explanatory content is what is invariant across equivalent configurations, just as the structural realist claims that scientific knowledge survives theory change only in structural form. The connection is suggestive; a rigorous mapping would require engaging with the ontological commitments of structural realism, which our descriptive framework does not address. Dropping decisiveness—accepting that the explanation cannot resolve all distinctions—corresponds to van Fraassen's [22] constructive empiricist stance that science aims for empirical adequacy, not full theoretical truth.

The result shares structural features with Arrow's impossibility theorem [23]: three individually reasonable desiderata that are pairwise satisfiable but jointly impossible under a natural background condition. Like Arrow's theorem, the proof is short once the axioms are identified; the contribution lies in identifying the right axiomatic framework. Okasha [24] connected theory choice to Arrow's theorem; our result provides the complementary connection for underdetermination of explanation.

6.2 Regulatory Analysis

OCC SR 11-7 [25], EU AI Act Article 13 [26], and NIST AI RMF [27] each implicitly assume faithfulness, stability, and decisiveness are jointly achievable; Theorem 1 shows they are not. Compliant documentation should state which property is prioritized and report explanation variance. Actionable guidance is in *SI Appendix*, Table S2.

6.3 AI Safety

If a safety argument depends on a specific circuit being THE mechanism of a capability, the impossibility requires accounting for alternative circuits with potentially different safety properties. The 85 valid circuits found by [19] for a simple XOR task suggest real-world capabilities will admit far more. Safety arguments should target circuit-invariant properties.

6.4 The "Obvious" Objection

A natural objection is that the impossibility is intuitively obvious: if equivalent models disagree on explanations, no single explanation can agree with all of them. We note three responses. First, the axiom substitution analysis shows that the obvious version of the result (with a stronger decisiveness axiom) collapses to a dilemma—only the precisely calibrated definitions yield a genuine trilemma. Second, the necessity result shows the impossibility is sharp: it holds if and only if the Rashomon property obtains. Third, the resolution (G-invariant aggregation) and its connection to invariant decision theory provide constructive content that goes beyond the impossibility alone.

6.5 Closing

The impossibility does not counsel despair but precision: state which property is sacrificed, aggregate optimally, and report uncertainty. The practitioner's task is to diagnose whether the Rashomon property obtains and to communicate the tradeoff honestly.

Materials and Methods

The Lean 4 formalization (74 files, 349 theorems, 72 axioms, 0 `sorry`) is available at `https://github.com/DrakeCaraker/universal-explanation-impossibility`. Experimental code and data are in *SI Appendix*. The core impossibility theorem (`explanation_impossibility`) depends on zero axioms beyond the Rashomon property. Full axiom inventory and theorem list are in *SI Appendix*, Table S1.

Acknowledgments

[To be added.]

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